

AgriSpectra-Q — Complete Results and Images

Live Matrix, Frozen Scientific Benchmark, and Reproducibility Package

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1 Executive Summary

This PDF consolidates the readable results, tables, charts, maps, and file inventory from the AgriSpectra-Q project. It contains a live, georeferenced spectral-anomaly Matrix run on three real EnMAP GeoTIFF scenes and the separate frozen six-model scientific benchmark. The live analysis is an unsupervised spectral-anomaly proxy. It is not a disease or pest diagnosis and does not independently establish AgriSpectra-Q superiority over HSI-RF.

The frozen benchmark remains: AgriSpectra-Q mean F1 0.963985 versus HSI-RF 0.963141, with paired 95% CI $[-0.001205, +0.002680]$. Statistical superiority was not established. The operational reference remains HSI-RF, while AgriSpectra-Q is a promising but unproven hybrid research layer.

2 Live Matrix Results

The live run ID is AGRQ-LIVE-20260916-132530-587fc9. It processed actual EnMAP GeoTIFF data using windowed reading, 224 bands, 30 m resolution, and source CRS/GeoTransform metadata. Priority thresholds are scene-relative percentiles: P50, P80, and P95. They are prioritisation thresholds rather than disease thresholds.

Scene	Dimensions	Bands	Valid pixels	NoData	CRS	Zones
scene_01_DT0000205230	1153 × 1198	224	1,028,176	25.56%	EPSG:32753	407
scene_02	1210 × 1244	224	1,006,261	33.15%	EPSG:32645	864
scene_03	1152 × 1214	224	1,047,911	25.07%	EPSG:32636	438

The operational recommendation is: “Spectral-stress evidence detected. Prioritise field inspection to determine the underlying cause. Field verification required.” No pesticide, irrigation, fertiliser, or crop-removal recommendation is made.

3 Live Scene Metadata

3.1 scene_01_DT0000205230

Dimensions: 1153 × 1198; bands: 224; resolution: 30.0 m; valid pixels: 1,028,176; NoData: 25.56%; CRS: EPSG:32753; processing time: 37.14 seconds.

Thresholds: P50 = 0.7470; P80 = 1.1317; P95 = 1.8809; extracted high-priority connected zones = 407.

3.2 scene_02

Dimensions: 1210 × 1244; bands: 224; resolution: 30.0 m; valid pixels: 1,006,261; NoData: 33.15%; CRS: EPSG:32645; processing time: 111.41 seconds.

Thresholds: P50 = 0.6622; P80 = 1.1688; P95 = 1.7142; extracted high-priority connected zones = 864.

3.3 scene_03

Dimensions: 1152 × 1214; bands: 224; resolution: 30.0 m; valid pixels: 1,047,911; NoData: 25.07%; CRS: EPSG:32636; processing time: 49.11 seconds.

Thresholds: P50 = 0.6929; P80 = 1.0800; P95 = 1.6642; extracted high-priority connected zones = 438.

4 Frozen Scientific Benchmark

Model	Mean F1	PR-AUC	ROC-AUC	Interpretation
AgriSpectra-Q, 0.963985, 0.994732, 0.998665, Highest numerical mean F1	Adaptive Classical, 0.963396, 0.994684, 0.998641, Best calibration	HSI-RF, 0.963141, 0.994945, 0.998742, Strongest ranking baseline	48-band XGBoost, 0.952233, 0.992257, 0.998001, Classical baseline	Spectral XGBoost, 0.947784, 0.992399, 0.998057, Classical baseline
Current Hybrid, 0.899766, 0.961551, 0.987517, Weaker hybrid reference				

The benchmark used three real EnMAP scenes, five seeds, spatially grouped splits, validation-only calibration, grouped OOF residual methodology, frozen test predictions, and 20,000 paired bootstrap replicates. The target is a spectral anomaly proxy, not a field-confirmed disease or pest label.

5 Decision Intelligence

The constrained inspection experiment found near-ties between AgriSpectra-Q and HSI-RF. At 5% budget both achieved 24.49% proxy recall; at 10% both achieved 49.19%; at 20% HSI-RF achieved 92.33% and AgriSpectra-Q 92.22%. The current evidence does not support replacing HSI-RF.

Unique hard-decision corrections were 29 cases where Q corrected an RF error and 23 cases where RF was correct and Q was wrong, for a net +6 corrections. This is weak complementarity, not a systematic operational advantage.

6 Visual Results

The following pages contain the generated benchmark figures, calibration plots, inspection curves, decision-value plots, routing plots, and actual scene priority maps.

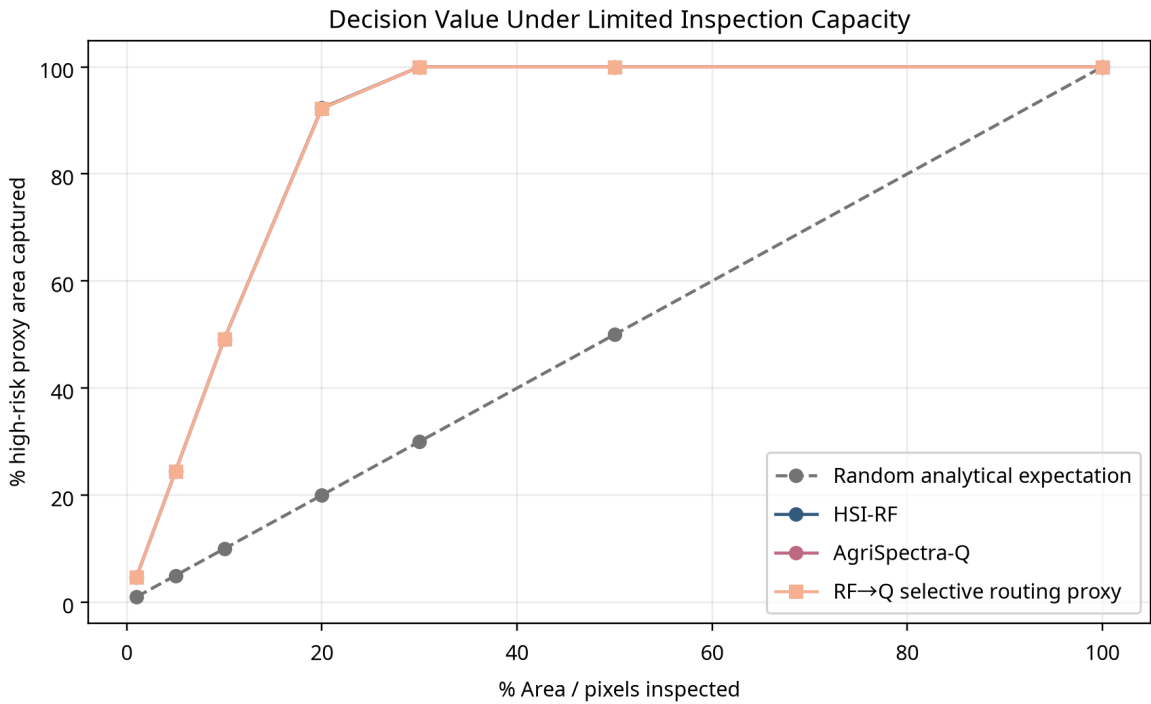


图 1 Decision Value Under Limited Inspection

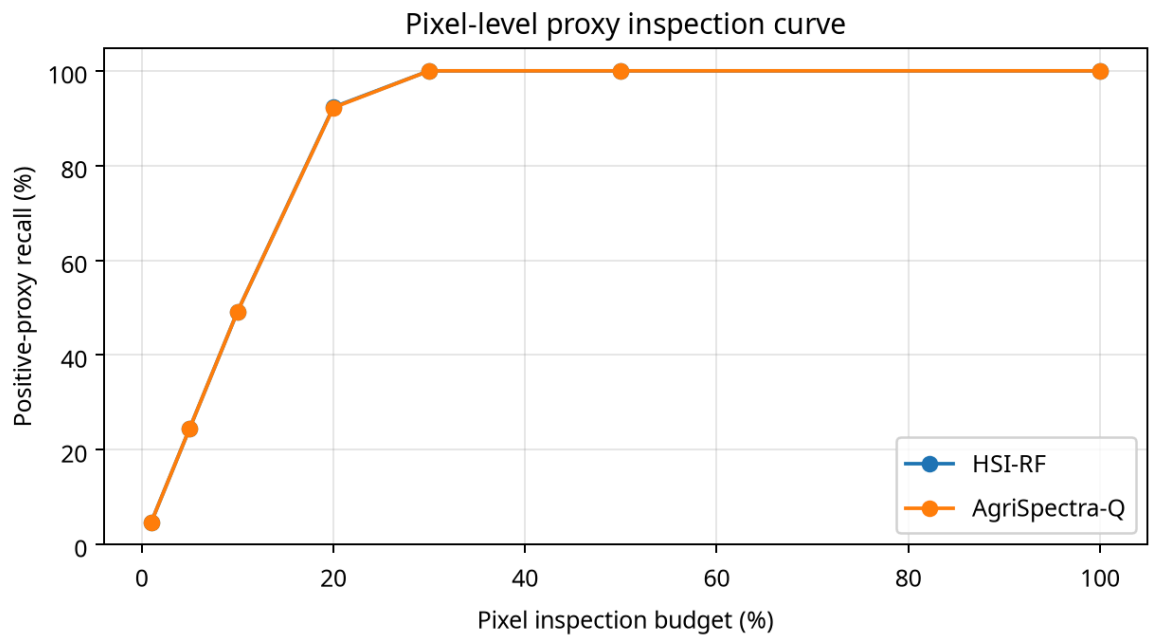


图 2 Inspection Recall Curve

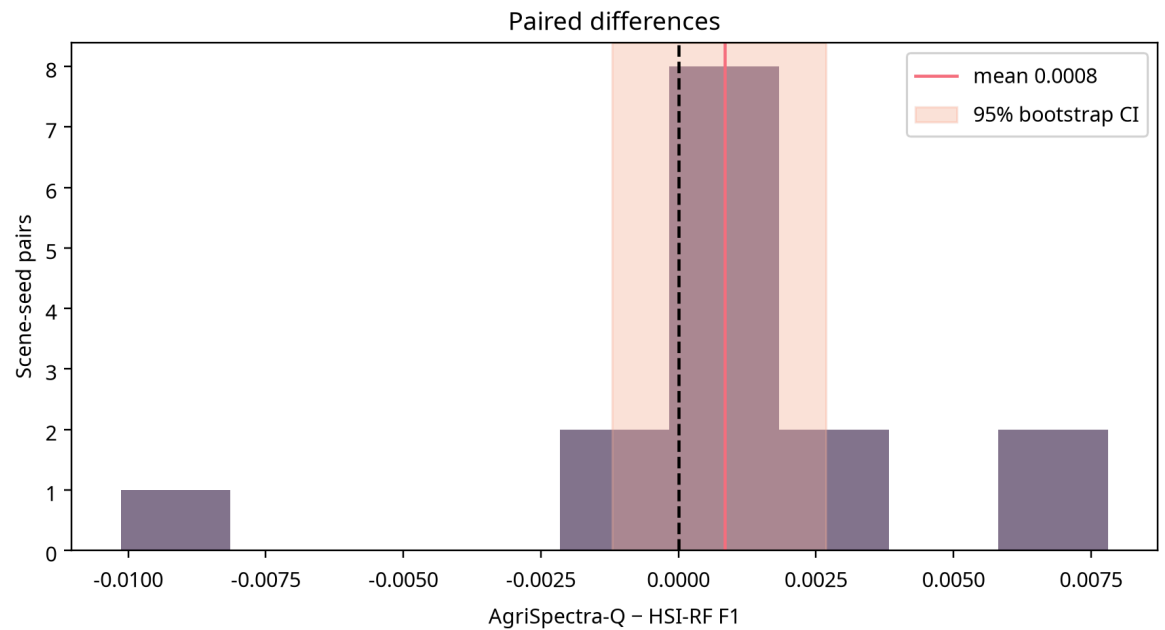


图 3 Paired Bootstrap

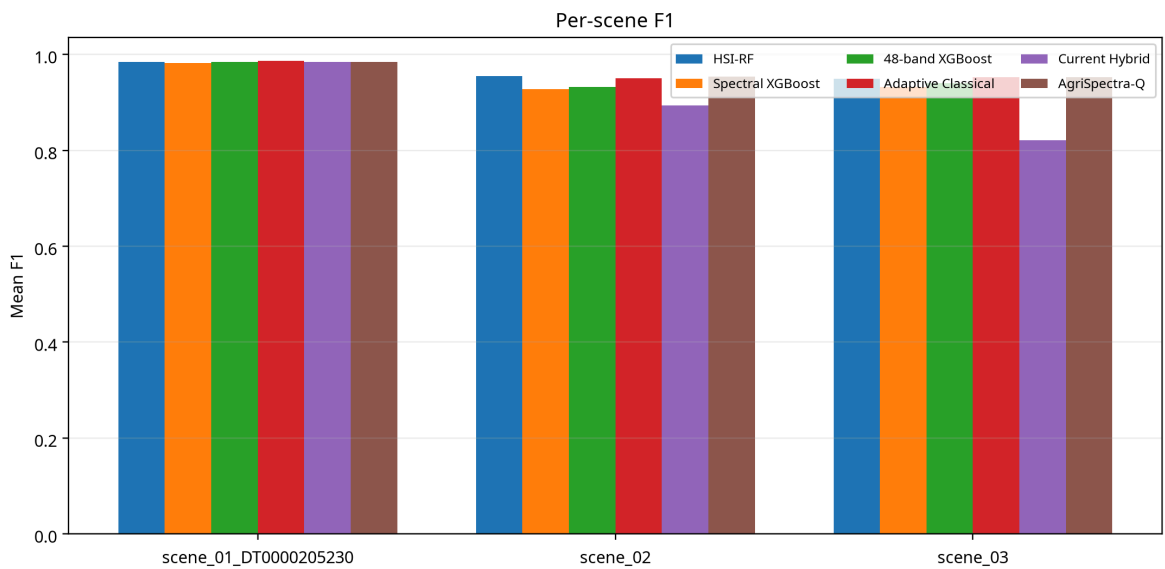


图 4 Per Scene F1

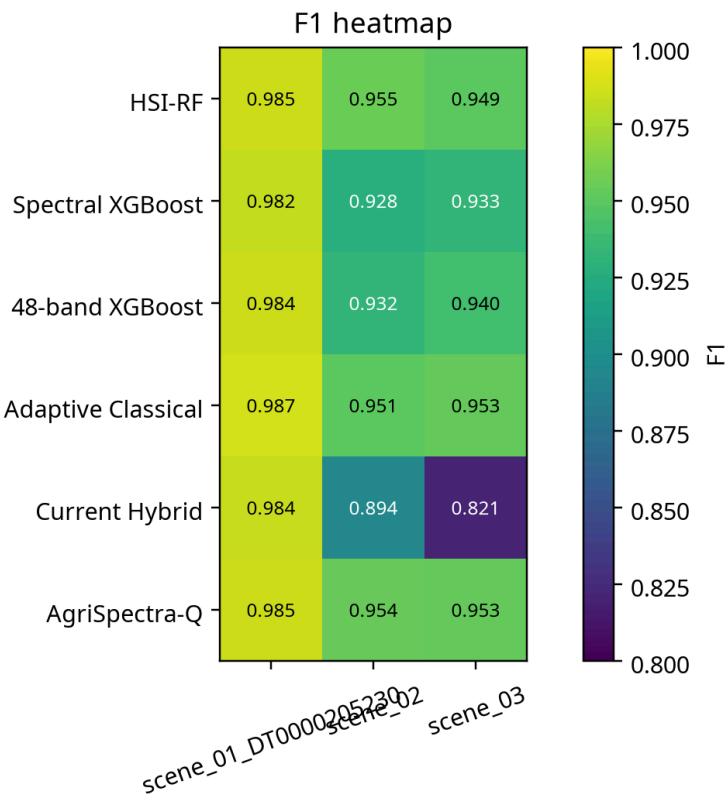


图 5 Per Scene Heatmap

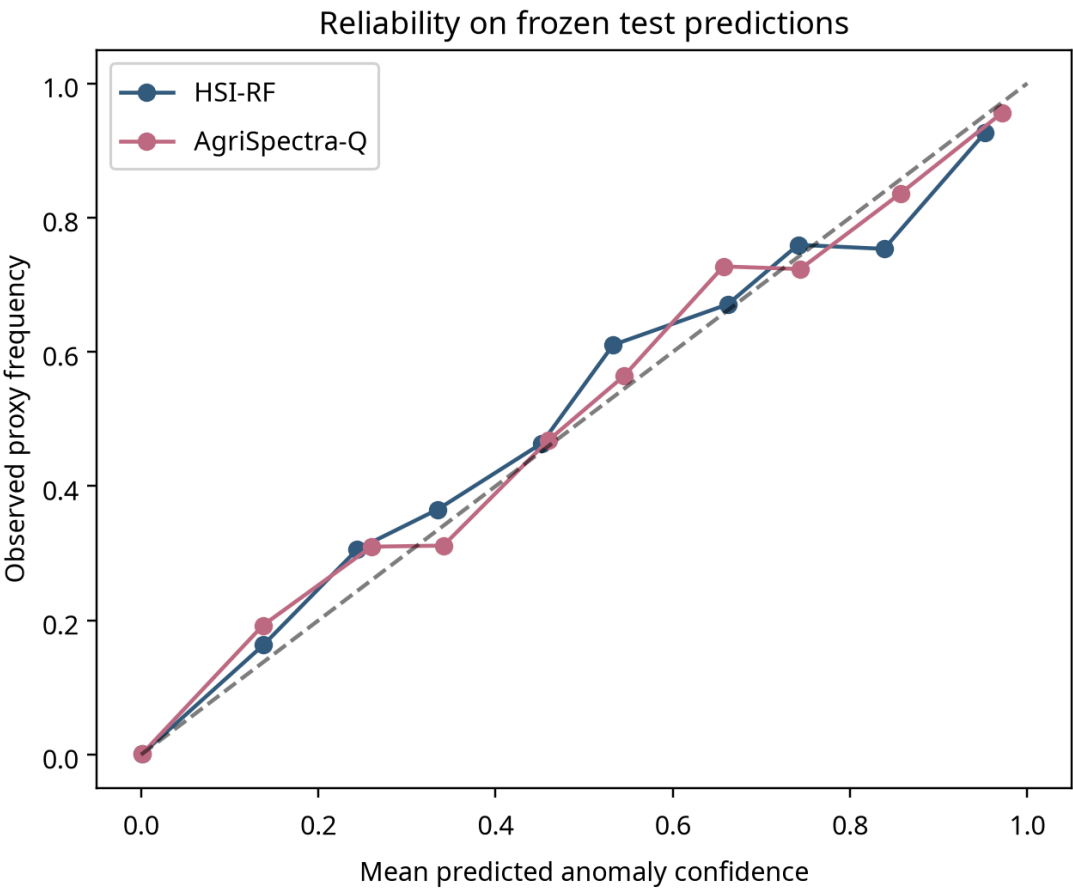


图 6 Reliability

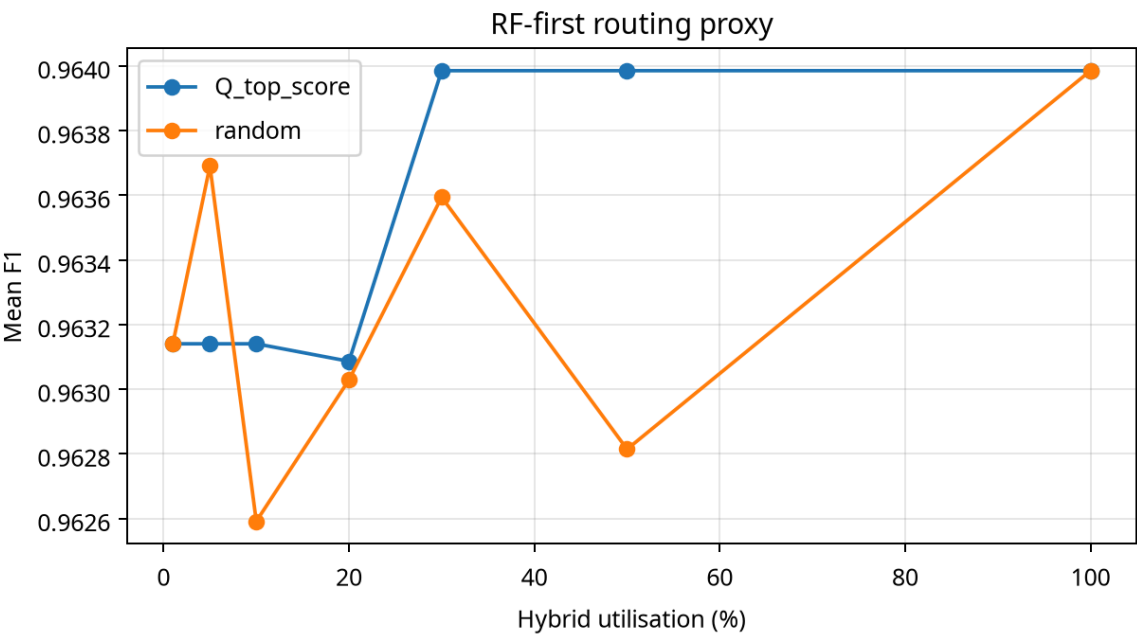


图 7 Routing F1 Curve

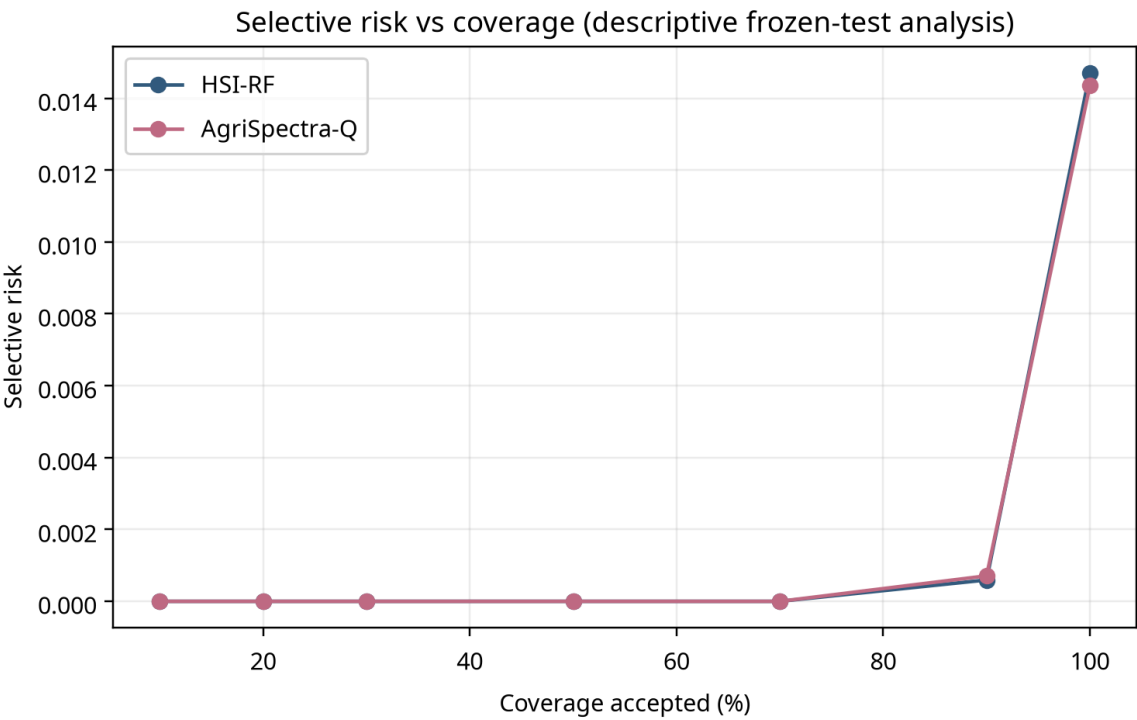


图 8 Selective Risk Curve

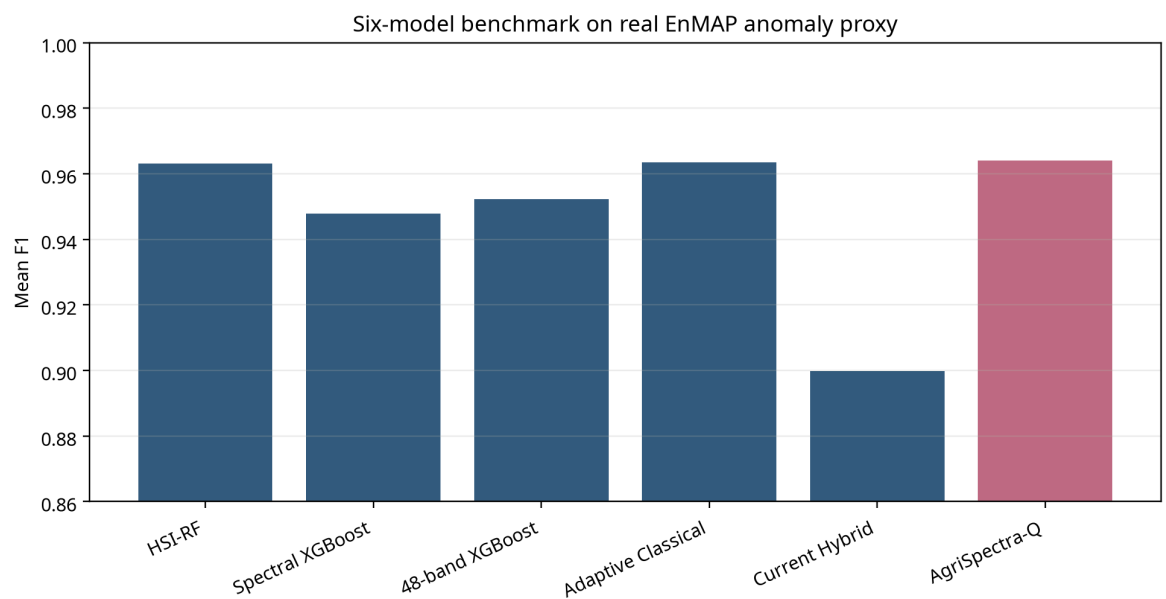


图 9 Six Model Benchmark



图 10 System Architecture

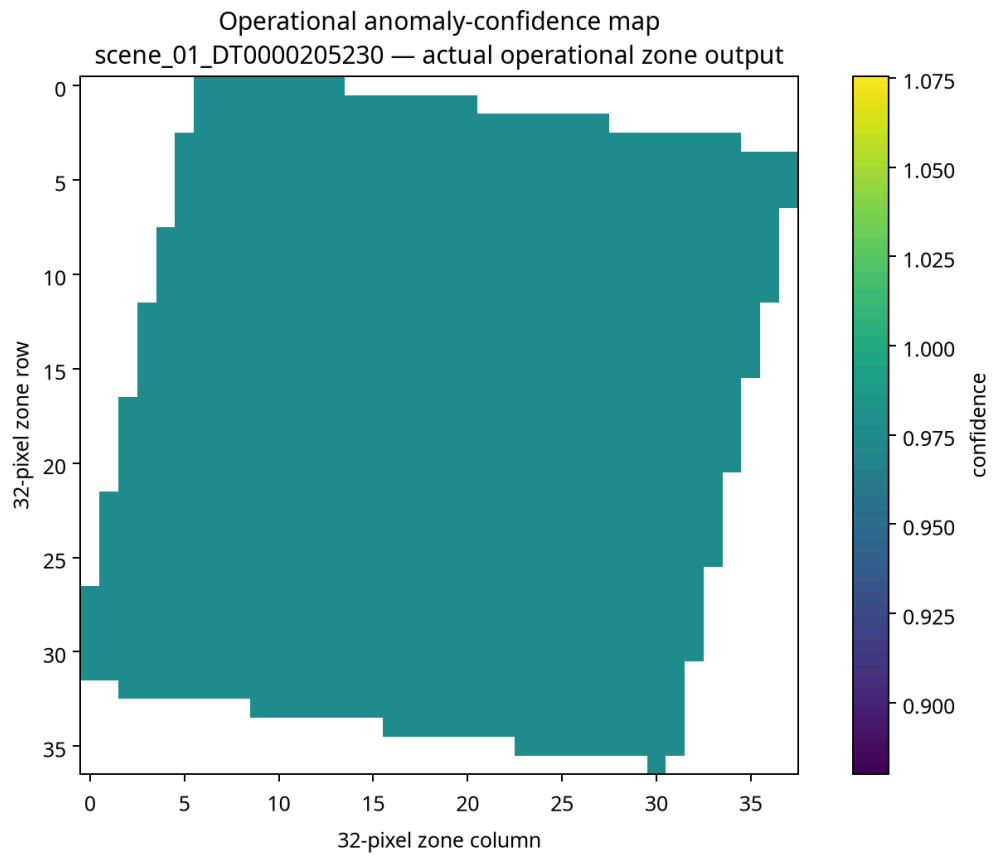


图 11 Scene 01 Dt0000205230 Confidence

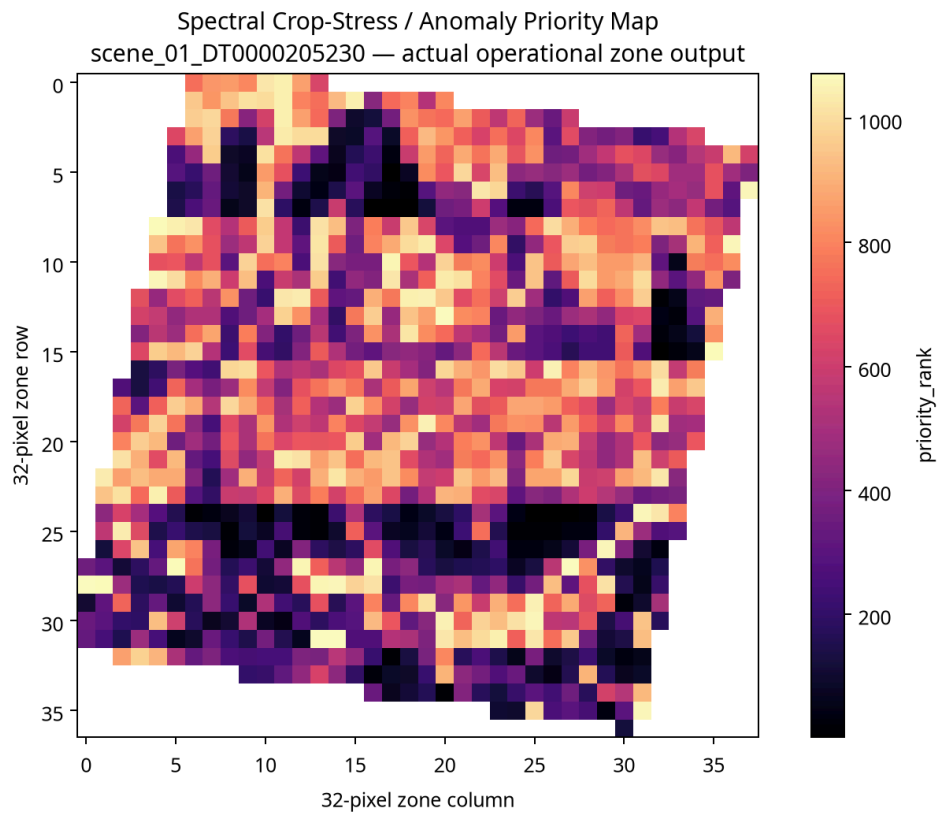


图 12 Scene 01 Dt0000205230 Priority Rank

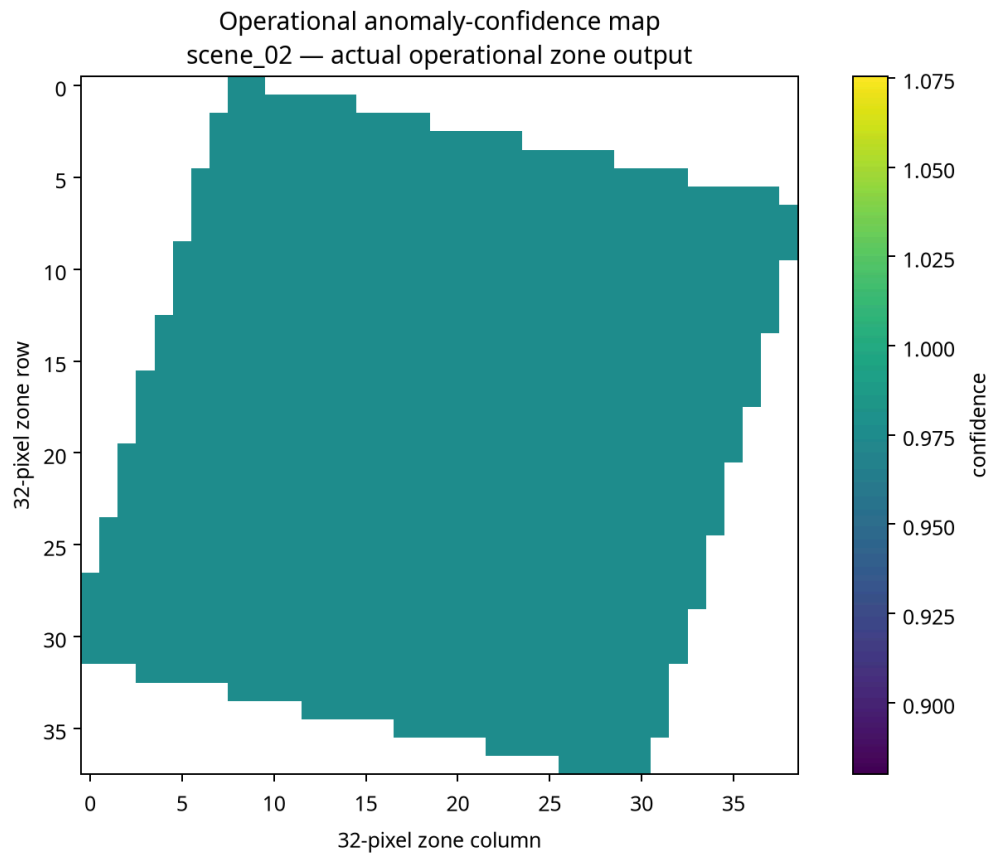


图 13 Scene 02 Confidence

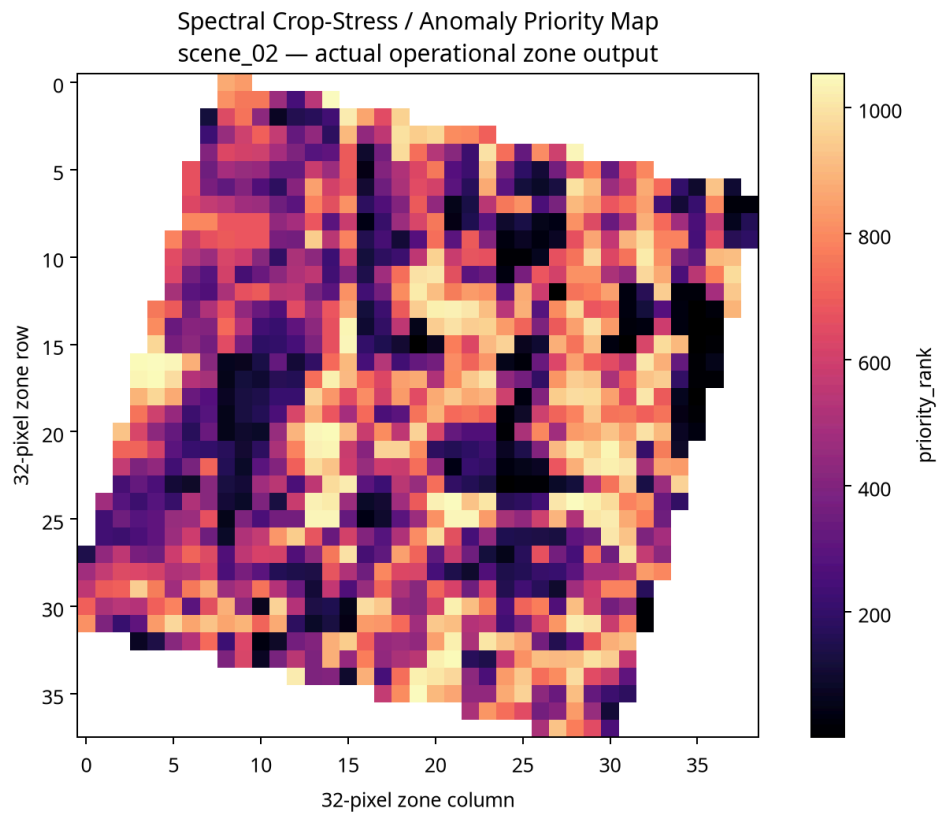


图 14 Scene 02 Priority Rank

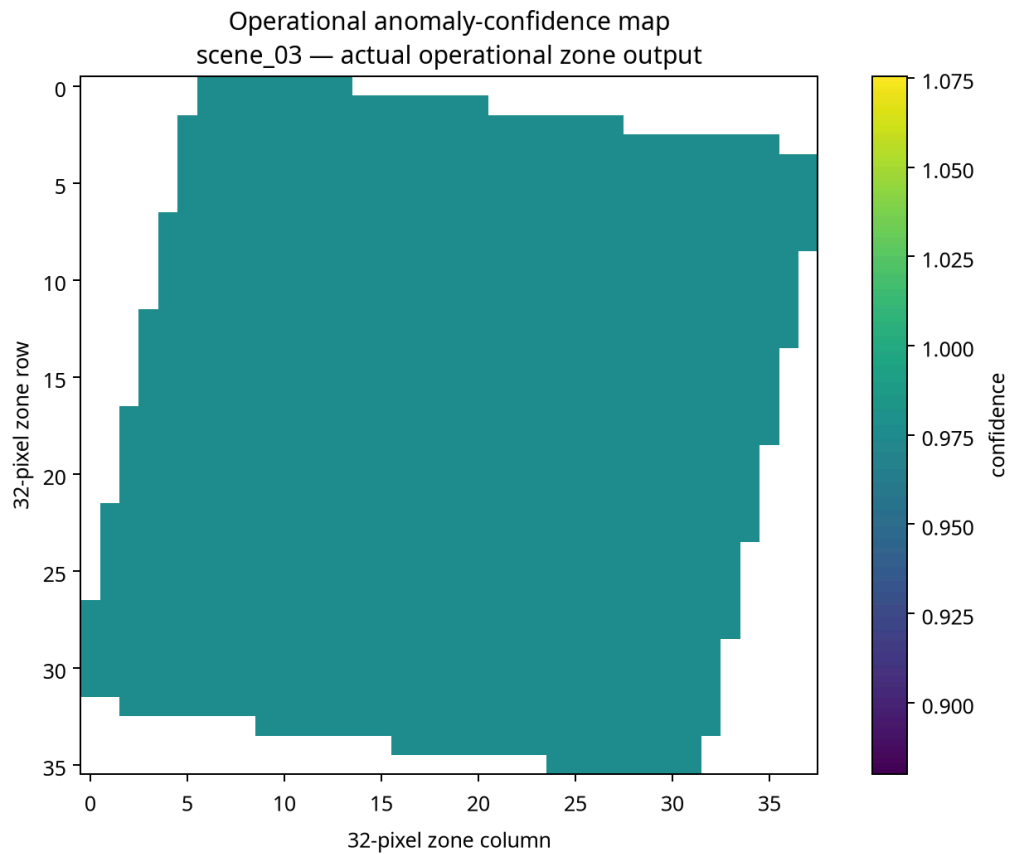


图 15 Scene 03 Confidence

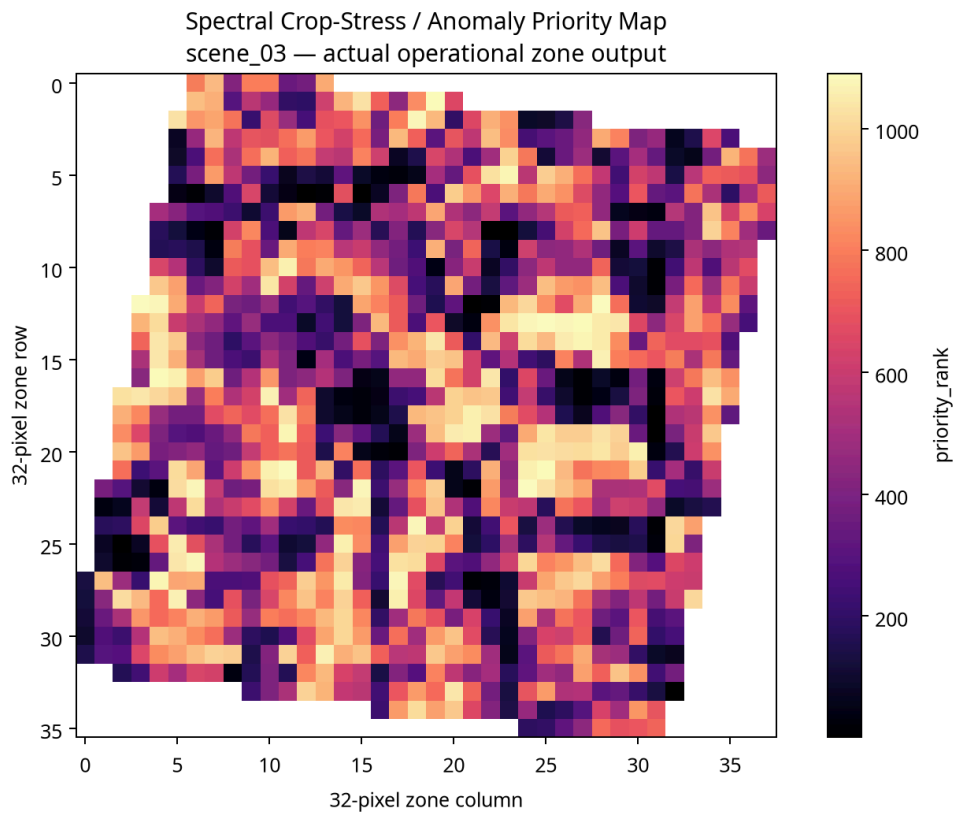


图 16 Scene 03 Priority Rank

7 File Inventory and Binary Artifacts

The accompanying ZIP package contains every result file and image. The PDF embeds all readable PNG figures and maps and summarizes all CSV/JSON outputs. Binary geospatial rasters, GeoJSON geometries, source code, and CSV files are preserved in their original form in the ZIP because converting them into PDF pages would destroy their machine-readable or geospatial properties.

The live run includes, for each scene: `risk_map.tif`, `priority_map.tif`, `zones.csv`, `zones.geojson`, `spectral_evidence.csv`, `inspection_budget.csv`, `scene_statistics.json`, `metrics.json`, and `manifest.json`. The industrial validation directory includes the final report, benchmark metrics, Decision Intelligence tables, Selective Risk, VOI proxy, quantum ablation, figures, dashboard, and manifests.

7.1 Complete file list

- `results/RESULTS_AND_IMAGES_README.md`
- `results/adaptive_hybrid/ablation_metrics.csv`
- `results/adaptive_hybrid/ablation_summary.csv`
- `results/adaptive_hybrid/adaptive_hybrid_final_report.md`
- `results/adaptive_hybrid/decision_cost_metrics.csv`
- `results/adaptive_hybrid/decision_cost_summary.csv`
- `results/adaptive_hybrid/forensics/band_distributions.json`
- `results/adaptive_hybrid/forensics/scene_distances.csv`
- `results/adaptive_hybrid/forensics/scene_stats.csv`
- `results/adaptive_hybrid/losc_metrics.csv`
- `results/adaptive_hybrid/losc_summary.csv`
- `results/adaptive_hybrid/operational/scene_01_DT0000205230_zone_risk.csv`
- `results/adaptive_hybrid/operational/scene_02_zone_risk.csv`
- `results/adaptive_hybrid/operational/scene_03_zone_risk.csv`
- `results/adaptive_hybrid/paired_bootstrap.json`
- `results/adaptive_hybrid/robustness_metrics.csv`
- `results/adaptive_hybrid/robustness_summary.csv`
- `results/enmap_20260707/enmap_validation_report.md`
- `results/enmap_20260707/manifest.json`
- `results/enmap_20260707/metrics.csv`
- `results/enmap_20260707/summary.csv`
- `results/enmap_three_scenes/aggregate_summary.csv`
- `results/enmap_three_scenes/all_metrics.csv`
- `results/enmap_three_scenes/all_summaries.csv`
- `results/enmap_three_scenes/scene_01_DT0000205230/manifest.json`
- `results/enmap_three_scenes/scene_01_DT0000205230/metrics.csv`
- `results/enmap_three_scenes/scene_01_DT0000205230/summary.csv`
- `results/enmap_three_scenes/scene_02/manifest.json`
- `results/enmap_three_scenes/scene_02/metrics.csv`
- `results/enmap_three_scenes/scene_02/summary.csv`
- `results/enmap_three_scenes/scene_03/manifest.json`

- results/enmap_three_scenes/scene_03/metrics.csv
- results/enmap_three_scenes/scene_03/summary.csv
- results/enmap_three_scenes/three_scene_report.md
- results/figures/baseline_comparison.png
- results/figures/forensic_comparison.png
- results/figures/risk_map.png
- results/figures/spectral_signature.png
- results/final_six_benchmark/error_analysis.csv
- results/final_six_benchmark/final_6_model_benchmark.csv
- results/final_six_benchmark/final_6_model_manifest.json
- results/final_six_benchmark/final_6_model_report.md
- results/final_six_benchmark/final_6_model_summary.csv
- results/final_six_benchmark/paired_bootstrap.json
- results/final_six_benchmark/predictions.json
- results/final_six_benchmark/robustness.csv
- results/final_six_benchmark/verified_paired_bootstrap.json
- results/final_six_benchmark/verified_summary.csv
- results/industrial_validation/ablation_results.csv
- results/industrial_validation/calibration_results.csv
- results/industrial_validation/computational_cost.csv
- results/industrial_validation/cost_sensitive_results.csv
- results/industrial_validation/dashboard/data.js
- results/industrial_validation/dashboard/index.html
- results/industrial_validation/decision_cards.csv
- results/industrial_validation/decision_value_results.csv
- results/industrial_validation/difficulty_quantile_results.csv
- results/industrial_validation/error_disagreement_analysis.csv
- results/industrial_validation/experiment_manifest.json
- results/industrial_validation/figures/decision_value_under_limited_inspection.png
- results/industrial_validation/figures/inspection_recall_curve.png
- results/industrial_validation/figures/paired_bootstrap.png
- results/industrial_validation/figures/per_scene_f1.png
- results/industrial_validation/figures/per_scene_heatmap.png
- results/industrial_validation/figures/reliability.png
- results/industrial_validation/figures/routing_f1_curve.png
- results/industrial_validation/figures/selective_risk_curve.png
- results/industrial_validation/figures/six_model_benchmark.png
- results/industrial_validation/figures/system_architecture.png
- results/industrial_validation/final_industrial_validation_report.md
- results/industrial_validation/final_industrial_validation_summary.csv
- results/industrial_validation/inspection_budget_results.csv
- results/industrial_validation/inspection_budget_summary.csv

- results/industrial_validation/inspection_recall_curve.png
- results/industrial_validation/loso_results.csv
- results/industrial_validation/main_metrics.csv
- results/industrial_validation/maps/scene_01_DT0000205230_confidence.png
- results/industrial_validation/maps/scene_01_DT0000205230_priority_rank.png
- results/industrial_validation/maps/scene_02_confidence.png
- results/industrial_validation/maps/scene_02_priority_rank.png
- results/industrial_validation/maps/scene_03_confidence.png
- results/industrial_validation/maps/scene_03_priority_rank.png
- results/industrial_validation/paired_bootstrap_results.json
- results/industrial_validation/quantum_component_ablation.csv
- results/industrial_validation/robustness_results.csv
- results/industrial_validation/routing_f1_curve.png
- results/industrial_validation/routing_results.csv
- results/industrial_validation/routing_summary.csv
- results/industrial_validation/selective_risk_results.csv
- results/industrial_validation/tables/table_01_six_model_benchmark.csv
- results/industrial_validation/tables/table_02_per_scene_results.csv
- results/industrial_validation/tables/table_03_statistical_comparison
- results/industrial_validation/tables/table_03_statistical_comparison.csv
- results/industrial_validation/tables/table_04_calibration
- results/industrial_validation/tables/table_05_error_disagreement
- results/industrial_validation/tables/table_06_inspection_budget
- results/industrial_validation/tables/table_07_cost_sensitive
- results/industrial_validation/tables/table_08_selective_routing
- results/industrial_validation/tables/table_09_ablation
- results/industrial_validation/tables/table_10_industrial_decision_matrix
- results/industrial_validation/tables/table_11_limitations_roadmap
- results/industrial_validation/tables/table_12_decision_value.csv
- results/industrial_validation/tables/table_13_unique_value.csv
- results/industrial_validation/tables/table_14_selective_risk.csv
- results/industrial_validation/tables/table_15_difficulty_quantiles.csv
- results/industrial_validation/tables/table_16_voi_proxy.csv
- results/industrial_validation/tables/table_17_quantum_component_ablation.csv
- results/industrial_validation/unique_corrections_by_scene.csv
- results/industrial_validation/unique_value_summary.csv
- results/industrial_validation/voi_proxy_results.csv
- results/live_matrix/AGRQ-LIVE-20260916-132530-587fc9/run_summary.json
- results/live_matrix/AGRQ-LIVE-20260916-132530-587fc9/
scene_01_DT0000205230/inspection_budget.csv
- results/live_matrix/AGRQ-LIVE-20260916-132530-587fc9/
scene_01_DT0000205230/manifest.json

- results/live_matrix/AGRQ-LIVE-20260916-132530-587fc9/scene_01_DT0000205230/metrics.json
- results/live_matrix/AGRQ-LIVE-20260916-132530-587fc9/
scene_01_DT0000205230/priority_map.tif
- results/live_matrix/AGRQ-LIVE-20260916-132530-587fc9/scene_01_DT0000205230/risk_map.tif
- results/live_matrix/AGRQ-LIVE-20260916-132530-587fc9/
scene_01_DT0000205230/scene_statistics.json
- results/live_matrix/AGRQ-LIVE-20260916-132530-587fc9/
scene_01_DT0000205230/spectral_evidence.csv
- results/live_matrix/AGRQ-LIVE-20260916-132530-587fc9/scene_01_DT0000205230/zones.csv
- results/live_matrix/AGRQ-LIVE-20260916-132530-587fc9/
scene_01_DT0000205230/zones.geojson
- results/live_matrix/AGRQ-LIVE-20260916-132530-587fc9/scene_02/inspection_budget.csv
- results/live_matrix/AGRQ-LIVE-20260916-132530-587fc9/scene_02/manifest.json
- results/live_matrix/AGRQ-LIVE-20260916-132530-587fc9/scene_02/metrics.json
- results/live_matrix/AGRQ-LIVE-20260916-132530-587fc9/scene_02/priority_map.tif
- results/live_matrix/AGRQ-LIVE-20260916-132530-587fc9/scene_02/risk_map.tif
- results/live_matrix/AGRQ-LIVE-20260916-132530-587fc9/scene_02/scene_statistics.json
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- results/live_matrix/AGRQ-LIVE-20260916-132530-587fc9/scene_02/zones.csv
- results/live_matrix/AGRQ-LIVE-20260916-132530-587fc9/scene_02/zones.geojson
- results/live_matrix/AGRQ-LIVE-20260916-132530-587fc9/scene_03/inspection_budget.csv
- results/live_matrix/AGRQ-LIVE-20260916-132530-587fc9/scene_03/manifest.json
- results/live_matrix/AGRQ-LIVE-20260916-132530-587fc9/scene_03/metrics.json
- results/live_matrix/AGRQ-LIVE-20260916-132530-587fc9/scene_03/priority_map.tif
- results/live_matrix/AGRQ-LIVE-20260916-132530-587fc9/scene_03/risk_map.tif
- results/live_matrix/AGRQ-LIVE-20260916-132530-587fc9/scene_03/scene_statistics.json
- results/live_matrix/AGRQ-LIVE-20260916-132530-587fc9/scene_03/spectral_evidence.csv
- results/live_matrix/AGRQ-LIVE-20260916-132530-587fc9/scene_03/zones.csv
- results/live_matrix/AGRQ-LIVE-20260916-132530-587fc9/scene_03/zones.geojson
- results/live_matrix/AGRQ-LIVE-API-ceb0d53f/run_summary.json
- results/live_matrix/AGRQ-LIVE-API-ceb0d53f/scene_03/inspection_budget.csv
- results/live_matrix/AGRQ-LIVE-API-ceb0d53f/scene_03/manifest.json
- results/live_matrix/AGRQ-LIVE-API-ceb0d53f/scene_03/metrics.json
- results/live_matrix/AGRQ-LIVE-API-ceb0d53f/scene_03/priority_map.tif
- results/live_matrix/AGRQ-LIVE-API-ceb0d53f/scene_03/risk_map.tif
- results/live_matrix/AGRQ-LIVE-API-ceb0d53f/scene_03/scene_statistics.json
- results/live_matrix/AGRQ-LIVE-API-ceb0d53f/scene_03/spectral_evidence.csv
- results/live_matrix/AGRQ-LIVE-API-ceb0d53f/scene_03/zones.csv
- results/live_matrix/AGRQ-LIVE-API-ceb0d53f/scene_03/zones.geojson
- results/locked_benchmark_v3/aggregate_summary.json
- results/locked_benchmark_v3/inspection_aggregate.json
- results/locked_benchmark_v3/inspection_budget_metrics.csv

- results/locked_benchmark_v3/locked_benchmark_v3_report.md
- results/locked_benchmark_v3/locked_manifest.json
- results/locked_benchmark_v3/locked_metrics.csv
- results/locked_benchmark_v3/locked_summary.csv
- results/locked_benchmark_v3/loso_aggregate.json
- results/locked_benchmark_v3/loso_metrics.csv
- results/locked_benchmark_v3/loso_summary.csv
- results/reports/execution_status.md
- results/reports/forensic_summary.md
- results/reports/multiseed_validation_report.md
- results/reports/s2_poc_report.json
- results/residual_v4/metrics.csv
- results/residual_v4/summary.csv
- results/residual_v4/v4_residual_report.md
- results/tables/ablation_results.csv
- results/tables/access_metadata.json
- results/tables/baseline_results.csv
- results/tables/cost_sensitive_results.csv
- results/tables/execution_manifest.json
- results/tables/forensic_manifest.json
- results/tables/forensic_metrics.csv
- results/tables/metrics.csv
- results/tables/multiseed_metrics.csv
- results/tables/multiseed_summary.csv
- results/tables/predictions.csv
- results/tables/statistical_results.json

8 Scientific Boundary

Live Matrix outputs are real georeferenced spectral-anomaly priority results. They do not represent confirmed disease, pest, field stress, financial return, or AgriSpectra-Q superiority over HSI-RF. The correct operational workflow is: Detect → Prioritise → Inspect → Verify.

9 References

[1]: https://www.enmap.org/data_access/ “EnMAP data access and mission information” [2]: <https://download.geoservice.dlr.de/ENMAP/files/> “EOC Geoservice EnMAP download service”